

Exam 2

個人成績報告

學生: 313554060 · 報告產生時間: 2026/06/14 13:06

滿分: 88.6 / 104 · 正確率: 82%

逐題總覽

題號	題型	狀態	分數
Q1	簡答題	✓	3 / 3
Q2	簡答題	✓	3 / 3
Q3	簡答題	✓	2 / 2
Q4	簡答題	△	2.5 / 3
Q5	簡答題	✓	3 / 3
Q6	簡答題	✓	2 / 2
Q7	簡答題	✓	4 / 4
Q8	簡答題	✓	3 / 3
Q9	簡答題	✓	2 / 2
Q10	簡答題	✓	2 / 2
Q11	簡答題	✓	2 / 2
Q12	簡答題	✓	2 / 2
Q13	簡答題	✓	3 / 3
Q14	簡答題	✓	4 / 4
Q15	簡答題	✓	2 / 2
Q16	簡答題	✓	3 / 3
Q17	簡答題	✓	3 / 3
Q18	簡答題	✓	4 / 4
Q19	簡答題	✓	2 / 2
Q20	單選題	✗	0→0 / 2.5
Q21	單選題	✗	0→0 / 2.5
Q22	單選題	✓	2→2.5 / 2.5
Q23	單選題	✓	2→2.5 / 2.5
Q24	單選題	✓	2→2.5 / 2.5
Q25	單選題	✓	2→2.5 / 2.5
Q26	單選題	✓	2→2.5 / 2.5
Q27	單選題	✗	0→0 / 2.5
Q28	單選題	✓	2→2.5 / 2.5
Q29	單選題	✗	0→0 / 2.5

題號	題型	狀態	分數
Q30	單選題	✓	2→2.5 / 2.5
Q31	單選題	✓	2→2.5 / 2.5
Q32	單選題	✓	2→2.5 / 2.5
Q33	單選題	✓	2→2.5 / 2.5
Q34	單選題	✓	2→2.5 / 2.5
Q35	單選題	✓	2→2.5 / 2.5
Q36	單選題	✓	2→2.5 / 2.5
Q37	單選題	✓	2→2.5 / 2.5
Q38	單選題	✗	0→0 / 2.5
Q39 (重配分)	單選題	→	— / —
Q40 (重配分)	單選題	→	— / —
Q41 (重配分)	單選題	→	— / —
Q42 (重配分)	單選題	→	— / —
Q43 (重配分)	單選題	→	— / —
Q44	單選題	✓	2→2.5 / 2.5
Q45	單選題	✗	0→0 / 2.5

題目詳情

Q1 簡答題 (3分)

✓ 正確 3 / 3

What are the three fundamental exceptions that can occur during message exchanges in a client-server model?

你的答案

LostReq:Lost of request message
LostResp:Lost of response message
SerCrash:沒有成功執行

參考答案

- (1) LostReq: The request message is lost.
- (2) LostResp: The response message is lost.
- (3) SerCrash: The server crashes during the execution of the request.

Q2 簡答題 (3分)

✓ 正確 3 / 3

Describe the three main types of redundancy used to achieve fault tolerance.

你的答案

information redundancy:增加額外的資訊以便偵錯、修復
time redundancy:透過額外的計算來避免錯誤
space redundancy:增加額外的resource來避免錯誤

參考答案

- (1) Information Redundancy: Adding extra bits (like Hamming code) for error detection/recovery.
- (2) Time Redundancy: Performing computations multiple times to recover from transient faults.
- (3) Space Redundancy: Adding extra physical components to handle hardware failures.

Q3 簡答題 (2分)

✓ 正確 2 / 2

Identify two major differences between a traditional Multiprocessor (MP) and a Chip-level Multiprocessor (CMP).

你的答案

共享L2 cache影響效能:某個greedy core占用大量cache，進而傷害其他core效能
Chip-level multiprocessor的CPU連結緊密:一旦部分元件壞掉可能讓多個CPU一起失效

參考答案

- (1) CMPs share an L2 cache, where a greedy core can negatively impact others.
- (2) Because CMP cores are tightly integrated on one chip, hardware failures are more likely to bring down multiple CPUs simultaneously.

Q4 簡答題 (3分)

△ 部分 2.5 / 3

How does affinity scheduling impact cache misses, TLB misses, and page faults in a multiprocessor system?

你的答案

affinity scheduling可能會降低TLB misses，因為它會讓threads盡量回到自己的CPU執行，TLB存在CPU裡
不會降低page faults，因為page存在RAM or backing storage，跟thread在哪個CPU執行無關
不會降低cache misses

參考答案

It reduces cache and TLB misses by keeping a thread on the same CPU, utilizing existing cached data and TLB entries. It has no effect on page faults, as physical RAM is shared among all CPUs.

Q5 簡答題 (3分)

✓ 正確 3 / 3

Define Access Transparency, Location Transparency, and Replication Transparency in distributed systems.

你的答案

access transparency:讓local和remote端資源可以用相同的操作方式存取

location transparency:存取資源時不需知道他的實際位置

replication transparency:使用多個副本提升系統的可靠性和效能，使用者和應用程式不需要知道副本的存在

參考答案

Access Transparency: Local and remote resources are accessed via identical operations.

Location Transparency: Resources are accessed without knowing their physical location.

Replication Transparency: Multiple resource instances are utilized for reliability without the user's knowledge.

TA 評語

All three correct: same ops; no need to know location; replicas for reliability, user/application unaware

Q6 簡答題 (2分)

✓ 正確 2 / 2

Compare the number of switches required for a crossbar network versus an Omega network for n CPUs and n memory modules.

你的答案

crossbar需要 n^2 個crosspoints

Omega network需要 $n/2 \log n$ 個switches($\log n$ 的底是2)

參考答案

A Crossbar network requires n^2 switches. An Omega network requires $(n/2) \log_2 n$ switches.

Q7 簡答題 (4分)

✓ 正確 4 / 4

List four methods to notify a receiver in a non-blocking message passing system that a message has arrived.

你的答案

- 1.當訊息抵達時interrupt通知receiver
- 2.The caller週期性poll buffer狀態，狀態改變就取出訊息
- 3.建立一個pop-up thread來處理抵達的訊息
- 4.直接在interrupt handler處理抵達active message

參考答案

- (1) Interrupt the receiver.
- (2) Periodic polling of the buffer status.
- (3) Creating a pop-up thread.
- (4) Processing an active message inside the interrupt handler.

Q8 簡答題 (3分)

✓ 正確 3 / 3

Compare Sender-Initiated (SI-LB) and Receiver-Initiated (RI-LB) Load Balancing algorithms. Which one maintains performance under heavy load?

你的答案

SI-LB:新的process會先被分配給建立他的node，若該node overloaded，則會繼續隨機探測其他node求助，若該node underloaded則會接收請求，否則探測到上限N
SI-LB在heavy load situation效能不好
RI-LB:process執行結束時，node會自動檢查自己是否underloaded，若是，則會主動探測其他node要工作，若被探測的node overloaded，會把他的一個process拿走，否則探測到上限N
RI-LB在heavy load situation不會讓他惡化

參考答案

SI-LB probes for underloaded nodes, while RI-LB probes for overloaded nodes. RI-LB maintains better performance under heavy load because busy nodes do not waste CPU cycles probing; only idle nodes actively seek work.

Q9 簡答題 (2分)

✓ 正確 2 / 2

While using the Test-and-Set-Lock (TSL) instruction for mutex locks in MPs may cause system bus contention, please suggest two ways to mitigate it.

你的答案

- 1.read before TSL
- 2.TSL with back off

參考答案

- (1) Read before TSL: The thread reads the lock variable normally (which can be cached). It only attempts TSL if the lock appears free.
- (2) TSL with back-off: If TSL fails, the thread waits for a defined delay before retrying, reducing continuous bus lock attempts.

Q10 簡答題 (2分)

✓ 正確 2 / 2

In multicasting systems, specify the meaning of 0-reliable and M-out-of-N-reliable semantics.

你的答案

- 0-reliable:不需要任何接收者回覆 eg.time signal generator
M-out-of-N-reliable:multicasting群組有N個成員，server期望有M個接收者回覆 eg.RAID

參考答案

- 0-reliable: The sender expects no response from any receiver.
M-out-of-N-reliable: The sender expects responses from exactly M receivers out of a group of N.

Q11 簡答題 (2分)

✓ 正確 2 / 2

Explain how a channel-based publish/subscribe system can be implemented using group communication.

你的答案

每個channel對應一個group，subscriber已加入group的方式訂閱，publisher發送訊息給group，group中的所有成員都能收到訊息

參考答案

Each channel is mapped to a multicast group. Subscribing equates to joining the group. Publishing to the channel translates to multicasting the event to all group members.

Q12 簡答題 (2分)

✓ 正確 2 / 2

In an SMP system, under what conditions is it more appropriate to use a spin lock versus a blocking lock?

你的答案

spin lock:thread會一直busy waiting直到shared resource可用，適合競爭者少、等待時間短
blocking lock:thread會放棄CPU，直到shared resource可用才被喚醒，適合競爭者多、等待時間長

參考答案

Spin lock: Appropriate when the expected wait time is very short or contention is low.
Blocking lock: Appropriate when the wait time is long or contention is high, allowing the CPU to be yielded.

Q13 簡答題 (3分)

✓ 正確 3 / 3

What are the three basic architectural approaches for Multiprocessor Operating Systems?

你的答案

- 1.每個CPU都有自己的OS
- 2.master-slave multiprocessors
- 3.symmetric multiprocessors

參考答案

- (1) Each CPU has its own Operating System.
- (2) Master-Slave Multiprocessors.
- (3) Symmetric Multiprocessors (SMP).

TA 評語

三點皆答對。

Q14 簡答題 (4分)

✓ 正確 4 / 4

List four common failures that may happen during a Remote Procedure Call (RPC).

你的答案

- 1.無法定位server
- 2.request message is lost
- 3.reply message is lost
- 4.server收到request後當機

參考答案

- (1) Client cannot locate the server.
- (2) Request message is lost.
- (3) Reply message is lost.
- (4) Server crashes after receiving the request.

TA 評語

4/4：四項核心失敗皆涵蓋。

Q15 簡答題 (2分)

✓ 正確 2 / 2

Explain the Cache Coherence problem and the basic rules a system must follow when a CPU wants to write a cached word.

你的答案

當某個CPU要write一個同時存在多個cache的word時，系統需要通知大家這write request。若某個cache他持有的是clean copy，則直接丟棄副本即可，若某個cache他持有的是dirty copy，需要先把這被修改過的copy寫回memory，才能繼續執行請求或直接把被修改的copy傳給要寫入的CPU

參考答案

When a CPU wants to write a word which is in more than one caches, the system must inform all caches of the write request. If a cache has a clean copy, it can just discard the copy. If a cache has a "dirty" (modified) copy, the cache must either write the modified copy back to the memory before the write request can proceed or transfer the modified copy to the writer directly.

State three major advantages of the Publish/Subscribe middleware model over Point-to-Point communication.

你的答案

- 1.scalability:加入新的subscriber不影響原本的subscriber和publisher
- 2.decoupling:subscriber和publisher不需要知道彼此的存在，更彈性也更好維護
- 3.many-to-many communication:允許多個subscriber溝通

參考答案

- (1) Scalability.
- (2) Decoupling of publishers and subscribers (they don't need to know each other).
- (3) Asynchronous communication (handling offline participants).

TA 評語

3 points: scalability 、 decoupling 、 many-to-many

Demonstrate how a counting semaphore initialized to N can be implemented using the Linda coordination language.

你的答案

initialization:執行out("sem")N次
wait():in("sem")
signal():out("sem")

參考答案

Initialization: execute out("sem") N times.
Wait(): execute in("sem").
Signal(): execute out("sem").

Q18 簡答題 (4分)

✓ 正確 4 / 4

Provide two advantages and two disadvantages of a Master-Slave architecture for multiprocessors.

你的答案

pros:

easier design and manage:

只有master processor需要OS和控制系統，設計管理較簡單

resource分配得更有效率:

master processor會把任務分配給slave processor，對平行處理有幫助，也有利load balancing

cons:

limited scalability:

系統高度依賴master processor，master processor容易形成performance bottleneck

single point of failure:

master processor一旦壞掉，整個系統會失效

參考答案

Pros: (1) Easier to design and manage. (2) Efficient resource allocation by a centralized master.

Cons: (1) Limited scalability (master becomes a bottleneck). (2) Single point of failure.

Q19 簡答題 (2分)

✓ 正確 2 / 2

Why are non-blocking operations less prone to deadlock?

你的答案

non-blocking 的其中一個優點就是系統不容易發生deadlock，因為他可以不讓deadlock的四個必要條件同時成立

參考答案

They allow processes to continue execution without waiting.

Q20 單選題 (2→2.5分)

✗ 錯誤 0→0 / 2.5

Gang scheduling in multiprocessor systems is primarily utilized to solve which of the following issues?

A. High page fault rates

B. TLB thrashing

C. Out-of-phase communication among related threads

正確

D. Excessive bus contention

你的選擇

你的答案: D · 參考答案: C

Q21 單選題 (2→2.5分)

✖ 錯誤 0→0 / 2.5

Which of the following interconnection network topologies stalls the entire chain of flits across multiple switches if the packet header cannot proceed due to a busy output channel?

A. Circuit Switching

你的選擇

B. Store and Forward Switching

C. Cut-Through Switching

D. Wormhole Routing

正確

你的答案: A · 參考答案: D

Q22 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

In an N-node hypercube interconnection network, the network diameter is proportional to:

A. $O(N)$

B. $O(N^2)$

C. $O(\sqrt{N})$

D. $O(\log_2(N))$

你的選擇 正確

你的答案: D · 參考答案: D

Q23 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

The concept of 'Bisection Width' in a multicomputer interconnection network evaluates:

A. The maximum number of hops between any two nodes.

B. The minimum number of links that must be failed to divide the network into two equal halves. 你的選擇 正確

C. The number of physical cables connected to the master node.

D. The size of the flit buffer in a network switch.

你的答案: B · 參考答案: B

Q24 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

A Beowulf Cluster is best described as:

- A. A tightly coupled Symmetric Multiprocessor (SMP).
- B. A chip-level multiprocessor with shared L2 cache.
- C. A distributed system utilizing the Andrew File System globally.
- D. A multicomputer built from commodity hardware, consisting of a master and slave nodes.

你的選擇 正確

你的答案: D · 參考答案: D

Q25 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

In the CBCAST (Causal Broadcast) protocol, which mechanism is fundamentally used to track and enforce the causal ordering of messages among a group of processes?

- A. A single, centralized global counter.
- B. Physical atomic hardware clocks.
- C. Vector timestamps (Vector clocks).
- D. Randomly generated sequence numbers.

你的選擇 正確

你的答案: C · 參考答案: C

Q26 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

Which of the following is the primary disadvantage of using variable-length messages instead of fixed-length messages in a message passing system?

- A. Variable-length messages cannot contain pointers.
- B. Variable-length messages consume significantly less bandwidth.
- C. They require complex dynamic memory allocation and garbage collection, causing overhead.
- D. They are fundamentally incompatible with Omega networks.

你的選擇 正確

你的答案: C · 參考答案: C

Q27 單選題 (2→2.5分)

✖ 錯誤 0→0 / 2.5

When an application continues to function normally despite the physical relocation of its required resources to a different server, the distributed system is exhibiting:

- | | |
|-----------------------------|------|
| A. Access Transparency | 你的選擇 |
| B. Migration Transparency | 正確 |
| C. Concurrency Transparency | |
| D. Failure Transparency | |

你的答案: A · 參考答案: B

Q28 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

In a multithreaded system, what is the primary advantage of a blocking receive primitive over a non-blocking receive primitive?

- | | |
|--|---------|
| A. It eliminates all network latency. | |
| B. It allows the thread to continuously poll the network buffer. | |
| C. It yields the CPU to other threads, resulting in better resource utilization. | 你的選擇 正確 |
| D. It prevents the cache coherence problem. | |

你的答案: C · 參考答案: C

Q29 單選題 (2→2.5分)

✖ 錯誤 0→0 / 2.5

Which fault tolerance technique involves executing a primary algorithm, checking the result with an Acceptance Test (AT), and rolling back to execute a secondary algorithm if the test fails?

- | | |
|------------------------------------|------|
| A. Triple Modular Redundancy (TMR) | 你的選擇 |
| B. Information Redundancy | |
| C. Recovery Blocks (RB) | 正確 |
| D. Byzantine Fault Tolerance | |

你的答案: A · 參考答案: C

Q30 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

What is the defining characteristic of a 'Transient' messaging system compared to a 'Persistent' messaging system?

- A. Transient systems store messages on disk until they are explicitly deleted by the user.
- B. Transient systems discard a message if it cannot be delivered while both the sender and receiver are actively running.
- C. Transient systems guarantee delivery even if the receiver's computer is powered off.
- D. Transient systems require the use of a shared mailbox.

你的選擇 正確

你的答案: B · 參考答案: B

Q31 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

Which type of Distributed System Middleware models the entire distributed environment as a hierarchical structure of directories and files?

- A. Document-based Middleware
- B. File System-based Middleware (e.g., Plan 9)
- C. Object-based Middleware (e.g., CORBA)
- D. Coordination-based Middleware (e.g., Linda)

你的選擇 正確

你的答案: B · 參考答案: B

Q32 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

When using the Direct Communication Model with Asymmetric Addressing, the communication primitives are defined such that:

- A. Both sender and receiver must name each other explicitly.
- B. Only the receiver must name the sender; the sender broadcasts to all.
- C. The sender names the receiver, but the receiver accepts messages from any process and returns the sender's ID.
- D. Neither party uses process IDs; they use shared mailboxes instead.

你的選擇 正確

你的答案: C · 參考答案: C

Q33 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

What is the key advantage of a crossbar switch?

- A. Low cost
- B. Non-blocking parallel access
- C. Minimal hardware
- D. Small size

你的選擇 正確

你的答案: B · 參考答案: B

Q34 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

In CC-NUMA systems, cache coherence is typically maintained using:

- A. Polling
- B. Paging
- C. Directory-based protocols
- D. Interrupts

你的選擇 正確

你的答案: C · 參考答案: C

Q35 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

Which statement about spinlocks is MOST accurate?

- A. They are always inefficient
- B. They are best for long critical sections
- C. They waste CPU cycles but avoid context switching
- D. They work only without cache

你的選擇 正確

你的答案: C · 參考答案: C

Q36 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

The key difference between MP and MC is:

- A. CPU speed
- B. Memory sharing
- C. Cache size
- D. OS type

你的選擇 正確

你的答案: B · 參考答案: B

Q37 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

Fat trees improve performance by:

- A. Reducing nodes
- B. Increasing CPU speed
- C. Increasing bandwidth near root
- D. Removing hierarchy

你的選擇 正確

你的答案: C · 參考答案: C

Q38 單選題 (2→2.5分)

✗ 錯誤 0→0 / 2.5

False sharing occurs when:

- A. Same variable shared
- B. Different data in same page
- C. Same CPU used
- D. Same OS

你的選擇

正確

你的答案: A · 參考答案: B

Q39 單選題 (2分 · 重配分)

→ 重配分 - / -

One-copy update semantics corresponds to:

- A. UNIX
- B. AFS
- C. HDFS
- D. NFS

正確

你的選擇

你的答案: C · 參考答案: A

Q40 單選題 (2分 · 重配分)

→ 重配分 - / -

Write-through caching is:

- A. Fast but unsafe
- B. Slow but reliable
- C. No disk usage
- D. Asynchronous

你的選擇 正確

你的答案: B · 參考答案: B

Q41 單選題 (2分 · 重配分)

→ 重配分 — / —

A replica differs from a cache because:

- A. It is temporary
- B. It is partial
- C. It is smaller
- D. It is persistent and known

你的選擇 正確

你的答案: D · 參考答案: D

Q42 單選題 (2分 · 重配分)

→ 重配分 — / —

Quorum consensus ensures consistency when:

- A. $r + w \leq n$
- B. $r + w > n$ and $2w > n$
- C. $r = w$
- D. $w = 1$

你的選擇

正確

你的答案: A · 參考答案: B

Q43 單選題 (2分 · 重配分)

→ 重配分 — / —

HDFS write policy is:

- A. Random write
- B. Write-many
- C. Read only
- D. Append only

你的選擇 正確

你的答案: D · 參考答案: D

Q44 單選題 (2→2.5分)

✓ 正確 2→2.5 / 2.5

At-least-once semantics may cause:

- A. No execution
- B. Duplicate execution
- C. Data loss
- D. Deadlock

你的選擇 正確

你的答案: B · 參考答案: B

Indirect communication uses:

A. Registers

B. Mailboxes

正確

C. CPU cache

你的選擇

D. Interrupts

你的答案: C · 參考答案: B